

This exam has 2 pages and 7 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. Equivalence relations (15 points)

Suppose that in the set $\mathbb{Z}^2$ (the set of all pairs of integer numbers), we define the following relation:

$$\langle a, b \rangle R \langle c, d \rangle \iff \text{either } a \leq d \text{ or } b < c \quad (\text{this is an } \textit{exclusive} \text{ or})$$

where $\leq$ and $<$ are the usual inequalities between integer numbers.

Determine whether the relation R is reflexive, symmetric, and/or transitive, and explain your answer in each case. Conclude from your answers whether or not the relation R is an equivalence relation.

2. Functions (6 + 4 + 5 points)

Consider the functions $abs: \mathbb{R} \rightarrow \mathbb{R}$ and $add_3: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$abs(x) = |x| \qquad add_3(x) = x + 3$$

- (a) For each function, determine whether or not it is injective and whether or not it is surjective. Please give a brief explanation of your answer in each case.
- (b) Give an explicit description (i.e. a formula for $f(x)$) of the function f defined by

$$f := add_3 \circ abs.$$

- (c) Express the function $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by the description

$$g(x) := |x - 6|$$

as a composition of the functions abs , add_3 , and their inverses.

3. Induction (5 + 10 points)

We claim that the following statement is true for all integer numbers $n \geq 0$:

$$\sum_{k=0}^n 2(k+1) = (n+1)(n+2)$$

(recall that $\sum_{k=0}^n a_k$ is shorthand notation for $a_0 + a_1 + a_2 + \dots + a_n$).

- (a) Verify by explicit calculation that the statement is true for $n = 0$, $n = 1$ and $n = 2$.
- (b) Using mathematical induction, prove that the statement is true for *all* integer numbers $n \geq 0$.

4. Equivalence classes for $\equiv$ (8 points)

Group the following ten formulas into their semantic equivalence classes:

$$\begin{array}{cccccc} \top & \perp & q & \neg p \vee q & \neg p \wedge q & p \rightarrow q \\ p \rightarrow (p \rightarrow q) & (p \rightarrow p) \rightarrow q & p \rightarrow (q \rightarrow q) & (p \rightarrow q) \rightarrow q & & \end{array}$$

5. Logic circuits (12 points)

Give a logic circuit corresponding to the formula $\phi \leftrightarrow \psi$, using only AND, OR and NOT gates.

6. OBDDs (9+4 points)

Let O_1 and O_2 be OBDDs, against the same variable order.

- (a) Explain how an OBDD for $O_1 \rightarrow O_2$ can be constructed from O_1 and O_2 .
- (b) Give an example to show that given reduced OBDDs O_1 and O_2 , the constructed OBDD for $O_1 \rightarrow O_2$ is not always reduced.

7. Predicate logic (3 + 3 + 6 points)

Consider the following two unary predicates:

- $C(x)$: x is a car
- $F(x)$: x has four wheels

- (a) Express the phrase “Every car has four wheels” in predicate logic.
Also express the phrase “If all elements are cars, then all elements have four wheels” in predicate logic. (“All elements” refers to the set A of the model.)
- (b) Explain why the first phrase semantically entails the second phrase.
- (c) Give a model to show that the second phrase does not semantically entail the first phrase.